FORENSIC TRIAGE REPORT — FORENSIGUARD

Timestamp (UTC): 2026-10-21T12:15:42.573794+00:00 | **Tool:** ForensiGuard v1.0.0

1. Evidence Provenance & Cryptographic Identifiers

Attribute	Value
Filename	src_0000_stego_high_payload.png
File Size	11,950 bytes
SHA-256	91dc99123c5b92e9465cfc0ab9448976...
MD5	826f24829dd1b3115f7e999f98b7ed70
Declared Ext / MIME	.png / image/png
Signature Check	Consistent

2. Probabilistic Model Inferences & Technique Attribution

Verdict	TAMPERED / MANIPULATED
Manipulation Probability	87.59%
Attributed Technique	STEGANOGRAPHY
Confidence Score	82.33%
Metadata XGBoost Score	0.37%
Image Residual CNN Score	51.01%
Clean-Baseline Anomaly	0.3301

3. Forensic Narrative & Explanations

• Time interval between modification and metadata change (mtime - ctime) (observed: -1.17, SHAP: -4.843) supported benign classification.

• Time interval between access and modification (atime - mtime) (observed: 182.17, SHAP: -0.499) supported benign classification.

• Isolation Forest deviation from clean filesystem baseline (observed: 0.33, SHAP: -0.025) supported benign classification.

• Time interval between access and metadata change (atime - ctime) (observed: 181.00, SHAP: -0.025) supported benign classification.

4. Legal Disclaimer & Evidentiary Scope

ForensiGuard provides probabilistic forensic triage indicators based on machine-learning models. These inferences do NOT constitute proof of guilt or conclusive evidentiary admissibility under Daubert/Frye. Independent manual examination of filesystem journal logs, file slack, and carving is mandatory.